

MARWAN EL SAABI GONZÁLEZ

DATA SCIENCE & MACHINE LEARNING | MSC BIOINFORMATICS CANDIDATE

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PROFILE

MSc student with hands-on experience across the data stack: time-series demand forecasting, training and benchmarking supervised ML models, building LLM-powered applications, and relational data modelling. Seeking a data science internship where I can turn analytical work into production-ready tools.

TECHNICAL PROJECTS & SOFTWARE DEVELOPMENT

Retail Demand Forecasting | Python, LightGBM, Pandas, Streamlit

- Forecast weekly SKU-level demand four weeks ahead from 1M+ real e-commerce transactions, using a global gradient-boosting model with lag, rolling-window, price and calendar features.
- Validated with rolling-origin backtesting against naive, seasonal-naive and moving-average baselines, improving wMAPE by 4.8% over the strongest baseline.

Alzheimer's Drug Classifier (ML Virtual Screening) | Python, Scikit-learn, XGBoost, RDKit

- Built a virtual screening pipeline on ChEMBL bioactivity data (IC50), encoding molecules as 1024-bit Morgan fingerprints with class-imbalance correction.
- Benchmarked Random Forest, SVM, XGBoost and KNN under stratified 5-fold cross-validation, validating the winning model on held-out compounds with leakage control.

AI-Driven Generative Application (De Novo Drug Designer) | Python, Streamlit, Google Gemini API, RDKit

- Architected and deployed an interactive open-source web application for *de novo* molecular generation and 3D rendering.
- Integrated the Gemini API with an RDKit validation layer to filter chemically invalid molecular structures.

RNA-seq Analysis Dashboard | Python, Streamlit, Plotly, Scikit-learn

- Built an interactive Streamlit dashboard for RNA-seq analysis: PCA, volcano/MA plots and heatmaps.
- Implemented differential expression analysis (Welch's t-test, Benjamini-Hochberg FDR correction) validated against a public dataset (Himes et al., 2014).

Melanoma Detection (Classical Computer Vision) | Python, OpenCV, Scikit-learn, Scikit-image

- Built a fully automatic pipeline segmenting dermoscopic lesions via an ensemble of Otsu, K-means and watershed, selected by IoU against manual ground truth.
- Extracted 40 shape, color and texture (LBP/GLCM) features; SVM classification under 5-fold cross-validation reached 90.2% accuracy and 84% melanoma sensitivity.

PROFESSIONAL EXPERIENCE

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| • Delivery Driver (Part-time) / Teigal Logística (Amazon Delivery Partner) | June 2026 to Present |
| • Production Operator / Stellantis España S.L, Vigo | Oct 2024 to Oct 2025 |
| • Seasonal Farmworker (Grape Harvest) / Iman Temporing ETT,S.L | Sept 2024 to Sept 2024 |

EDUCATION

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| University of A Coruña (Universidade da Coruña), A Coruña | Sept 2025 to Feb 2027 |
| MSc in Bioinformatics Applied to Health Sciences | |
| University of Vigo (Universidade de Vigo), Vigo | Sept 2019 to Jun 2023 |
| BSc in Biology (Specialization: Biotechnology) | |

AREAS OF EXPERTISE

- Programming & Data:** Python (NumPy, Pandas, OpenCV, Matplotlib), R (Tidyverse, ggplot2), SQL, MATLAB, Bash.
- Machine Learning:** Scikit-learn, XGBoost, LightGBM, Neural Networks, SVMs, Cross-Validation, Time-Series Forecasting.
- Tools & Infrastructure:** Linux, Git/GitHub, Docker, Conda, High-Performance Computing (SLURM).

LANGUAGES

English, Spanish, Galician